

Lab Instructions

Parallelize the following scenarios with multiprocessing

Scenario 1: AI Inference Serving

Users upload many images. We must classify them. Then, we use pre-trained scikit-learn model.

Dataset

```
from sklearn.datasets import load_digits  
  
X, y = load_digits(return_X_y=True)
```

Train

```
from sklearn.ensemble import RandomForestClassifier  
  
model = RandomForestClassifier()  
  
model.fit(X, y)
```

Sequential Serving

```
import time  
  
def predict_sample(sample):  
    time.sleep(0.05)  
    return model.predict(sample.reshape(1, -1))[0]  
  
predictions = []  
for sample in X[:200]:  
    predictions.append(  
        predict_sample(sample)  
    )
```

TODO:

- Turn the above sequential serving to parallel serving with multiprocessing
 - Measure the inference time of the serial version and parallel version.
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Scenario 2: Parallel Hyperparameter Search

Find the best decision tree.

Traditional search:

```
depths = [2,4,6,8,10,12]
```

Train each one using single process

```
from sklearn.tree import DecisionTreeClassifier  
  
from sklearn.model_selection import cross_val_score
```

```
def evaluate(depth):  
    model = DecisionTreeClassifier(  
        max_depth=depth  
    )
```

```
    score = cross_val_score(  
        model,  
        X,  
        y,  
        cv=5  
    ).mean()  
    return depth, score
```

```
results = []
```

```
for d in depths:
```

```
    results.append(evaluate(d))
```

TODO:

- Turn the above sequential search to parallel search with multiprocessing
- Measure the inference time of the serial version and parallel version.

Deliverables

Send your code to me by email, lidongvtemail@gmail.com
